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**Math 3108 – Linear Algebra**  
**Exam 3**  
**Prof. Runnion**  
**November 22, 2024**

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like  $\sqrt{2}$ ), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as  $\sqrt{9}$  and  $\cos \pi$  should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, **NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!**
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM  
UNTIL TOLD TO DO SO**

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**Multiple Choice and True/False.** Completely fill in the box corresponding to the correct answer(s). On the multiple choice problems, assume all matrices and vectors contain only real entries.

1. Consider the matrix  $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 3 & 7 & 9 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

(3 points) How many distinct eigenvalues does  $A$  have?

☐ 0   ☐ 1   ☐ 2   ☒ 3   ☐ 4   ☐ 5   ☐ 6

(3 points) Can you determine if  $A$  is diagonalizable without performing any computations?

☒ Yes   ☐ No

2. (3 points) If  $\lambda = 0$  is not an eigenvalue of  $A$ , then  $A$  is \_\_\_\_\_ invertible.

☒ a) always   ☐ b) sometimes   ☐ c) never

3. (3 points) The vector space  $\mathbb{R}^3$  is isomorphic with which of the following vector spaces?

☒ a)  $\mathbb{P}_2$    ☐ b)  $\mathbb{P}_3$    ☐ c)  $\mathbb{P}_4$

4. (3 points) An  $m \times n$  matrix  $U$  has \_\_\_\_\_ columns if and only if  $U^T U = I$ .

☐ a) Orthogonal   ☒ b) Orthonormal

5. (3 points) If  $A$  is a  $3 \times 7$  matrix, what is the smallest possible dimension of  $\text{Nul } A$ ?

☐ 0   ☐ 1   ☐ 2   ☐ 3   ☒ 4   ☐ 5   ☐ 6   ☐ 7   ☐ 8   ☐ 9

6. (3 points) If  $A$  is an  $n \times n$  matrix and  $\text{Col } A = \mathbb{R}^n$ , then  $A$  is \_\_\_\_\_ invertible.

☒ a) always   ☐ b) sometimes   ☐ c) never

7. (3 points) The rank of  $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$  is

☒ a) 2   ☐ b) 3   ☐ c) 4

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Open Answer. Show ALL your work!

8. (8 points) The set  $B = \{1+t^2, t-3t^2, 1+t-3t^2\}$  is a basis for  $\mathbb{P}_2$ . Find the polynomial  $q$

in  $\mathbb{P}_2$  given that  $[q]_B = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$ .

$$1+t^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$t-3t^2 = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -5 \end{bmatrix}$$

$$1+t-3t^2 = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$$

$$q = -1 + 2t - 5t^2$$

9. (8 points) The set  $B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}$  is a basis for  $\mathbb{R}^2$ . Find the coordinate vector  $[x]_B$  of

$$P_B[x]_B = \vec{x} \quad \vec{x} = \begin{bmatrix} 2 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} [x]_B = \begin{bmatrix} 2 \\ -6 \end{bmatrix}$$

$$[x]_B = \begin{bmatrix} -13 \\ 5 \end{bmatrix}$$

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10. a) (6 points) Find the distance between  $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$ .

$$\|\vec{u}-\vec{v}\| \quad \vec{u}-\vec{v} = \begin{bmatrix} 10 \\ 6 \\ -8 \end{bmatrix}$$

$$\sqrt{(10)^2 + (6)^2 + (-8)^2} = \sqrt{100 + 36 + 64} = \sqrt{200} = 10\sqrt{2}$$

distance =  $10\sqrt{2}$

- b) (3 points) Are  $\mathbf{u}$  and  $\mathbf{v}$  orthogonal?

$$24 - 9 - 15 = 0$$

$$\vec{u} \cdot \vec{v} = 0$$

☒ Yes

☐ No

(fill in the box)

11. (9 points) Find the (potentially complex) eigenvalues of  $A = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$

$$\det(A - \lambda I) = (3 - \lambda)(3 - \lambda) + 16 = 0$$

$$\lambda^2 - 6\lambda + 25 = 0$$

$$\lambda = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(25)}}{2(1)}$$

$$= \frac{6 \pm \sqrt{36 - 100}}{2}$$

$$= \frac{6 \pm 8i}{2}$$

$$\lambda = 3 - 4i, 3 + 4i$$

$\lambda =$   $3 - 4i, 3 + 4i$

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12. (12 points) The vectors  $\mathbf{x}_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$  and  $\mathbf{x}_2 = \begin{bmatrix} 7 \\ -7 \\ -4 \\ 1 \end{bmatrix}$  form a basis for a subspace  $W$  of  $\mathbb{R}^4$ .

Use the Gram-Schmidt process to produce an orthogonal basis  $\{\mathbf{v}_1, \mathbf{v}_2\}$  for  $W$

where  $\mathbf{x}_1 = \mathbf{v}_1$ .

$$\vec{v}_2 = \begin{bmatrix} 7 \\ -7 \\ -4 \\ 1 \end{bmatrix} - \frac{\vec{v}_1 \cdot \vec{x}_2}{\vec{x}_1 \cdot \vec{x}_1} \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ -7 \\ -4 \\ 1 \end{bmatrix} - \frac{36}{18} \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$$

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13. Consider the matrix  $A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ . The only eigenvalues of  $A$  are  $\lambda = 2$  and  $\lambda = 3$ .

a) (9 points) Find a basis for the eigenspace of  $A$  corresponding to  $\lambda = 2$ .

$$\begin{aligned} \lambda=2 \\ [A - 2I | \vec{0}] &= \begin{bmatrix} 0 & 0 & -2 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{swap } R_1, R_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} x_1 &= -x_2 \\ x_2 &\text{ is free} \\ x_3 &= 0 \end{aligned}$$

$$\vec{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

b) (9 points) Find a basis for the eigenspace of  $A$  corresponding to  $\lambda = 3$ .

$$\begin{aligned} \lambda=3 \\ [A - 3I | \vec{0}] &= \begin{bmatrix} -1 & 0 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} x_1 &= -2x_3 \\ x_2, x_3 &\text{ are free} \end{aligned}$$

$$\vec{x} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

c) (6 points) Diagonalize  $A$ . That is, find an invertible matrix  $P$  and a diagonal matrix  $D$  such that  $A = PDP^{-1}$ .

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

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14. (6 points) Find the matrix  $A$  of the quadratic form shown below. Assume  $\mathbf{x}$  is in  $\mathbb{R}^3$ .

$$x_1^2 + 3x_2^2 - 7x_3^2 + 2x_1x_2 - 4x_1x_3 + 6x_2x_3$$

Main diagonal

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 3 & 3 \\ -2 & 3 & -7 \end{bmatrix}$$

15.